

WT EXPERIMENT - 5

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AIM: To simulate the ultrasonic distance measurement system using Arduino UNO and HC-SR04 sensor in a virtual environment.

Theory:

Tinkercad is an online simulation tool that enables users to design, simulate, and test electronic circuits. It provides a virtual environment where components like Arduino, sensors, LEDs, LCDs, motors, and more can be connected and programmed without needing physical hardware.

Tinkercad is widely used for:

- Learning and testing Arduino projects.
- Simulating sensor-based applications.
- Debugging circuits before implementing them in real life.

In this project, we simulate distance measurement using an **Ultrasonic Sensor (HC-SR04)**. The ultrasonic sensor measures distance by emitting a high-frequency sound wave and timing how long it takes for the sound to reflect back from the object. The time taken is then used to calculate the distance.

The principle of operation involves sending an ultrasonic pulse from the **Trig** pin and receiving the echo on the **Echo** pin. The distance is calculated using the formula:

$$\text{Distance (cm)} = \frac{\text{Duration of Echo Pulse} \times \text{Speed of Sound (343 m/s)}}{2}$$
$$\text{Distance (cm)} = 2 \times \text{Duration of Echo Pulse} \times \text{Speed of Sound (343 m/s)}$$

Where:

- Speed of sound = 343 m/s or 0.0343 cm/ μ s
- Duration is the round-trip time of the pulse.

Objectives of Simulation

- To simulate the working of an ultrasonic distance sensor using Tinkercad.
- To measure the distance in real-time and display the results on an LCD screen.
- To debug and test the circuit before real-world implementation.

Steps:

Step 1: Open Tinkercad and Create a New Circuit

1. Go to Tinkercad (<https://www.tinkercad.com/>).
2. Click on "Circuits" from the left panel.
3. Click "Create New Circuit."

Step 2: Add Components

Search and drag the following components onto the workspace:

1. Arduino Uno
2. 16x2 LCD Display (without I2C)
3. HC-SR04 Ultrasonic Sensor
4. 10K Ω Potentiometer (for LCD contrast)
5. Resistors (220 Ω or 1K Ω for LCD backlight)
6. Jumper Wires

Step 3: Wire the Components

Wiring the LCD to Arduino

LCD Pin	Connect To
GND	Arduino GND
VCC	Arduino 5V
V0 (Contrast)	Middle Pin of Potentiometer
RS	Arduino Pin 7
RW	Arduino GND
E	Arduino Pin 8
D4	Arduino Pin 6
D5	Arduino Pin 5
D6	Arduino Pin 4
D7	Arduino Pin 3
A (Backlight +)	5V (via 220Ω resistor)

K (Backlight -) GND

Wiring the Ultrasonic Sensor (HC-SR04)

HC-SR04 Pin	Connect To
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VCC	Arduino 5V
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GND	Arduino GND
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Trig	Arduino Pin 9
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Echo	Arduino Pin 10
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Step 4: Write the Arduino Code

Click on "Code" and select "Text" mode.

Paste this code in the Arduino editor:

```
#include <LiquidCrystal.h>

// Initialize LCD (RS, E, D4, D5, D6, D7)
LiquidCrystal lcd(7, 8, 6, 5, 4, 3);

// Ultrasonic sensor pins
const int trigPin = 9;
const int echoPin = 10;

void setup() {
    lcd.begin(16, 2); // Initialize LCD
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    Serial.begin(9600); // Start serial monitor
}

void loop() {
    // Send a pulse
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
```

```
digitalWrite(trigPin, LOW);

// Read the echo
long duration = pulseIn(echoPin, HIGH);
int distance = duration * 0.0343 / 2; // Convert time to distance
(cm)

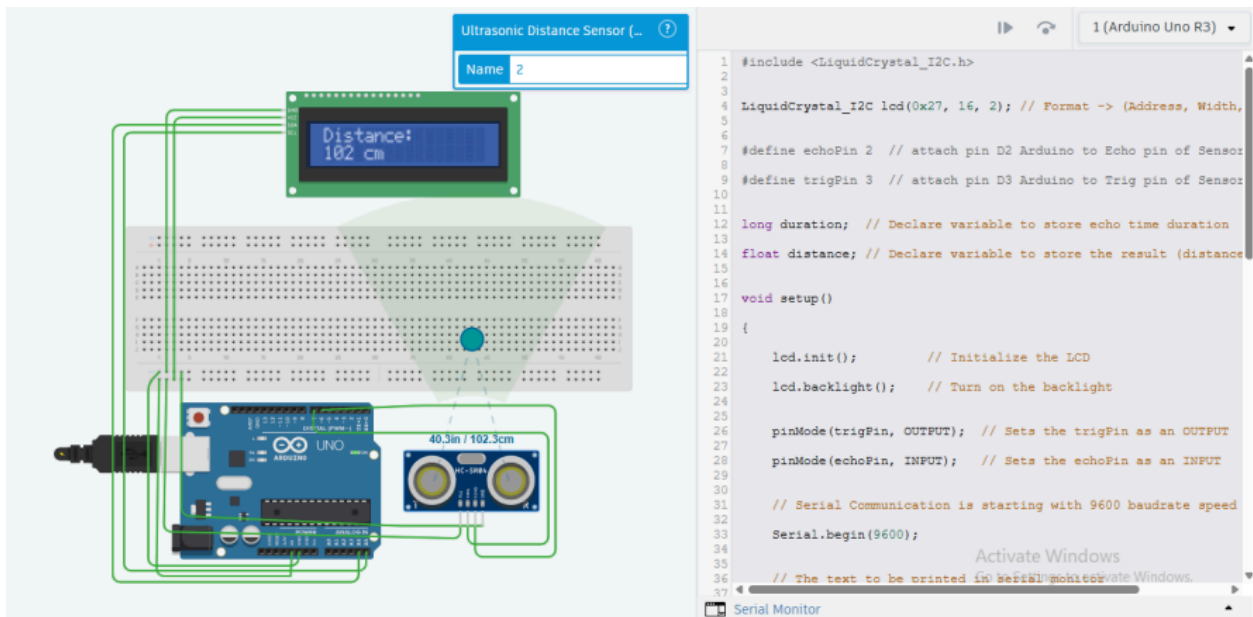
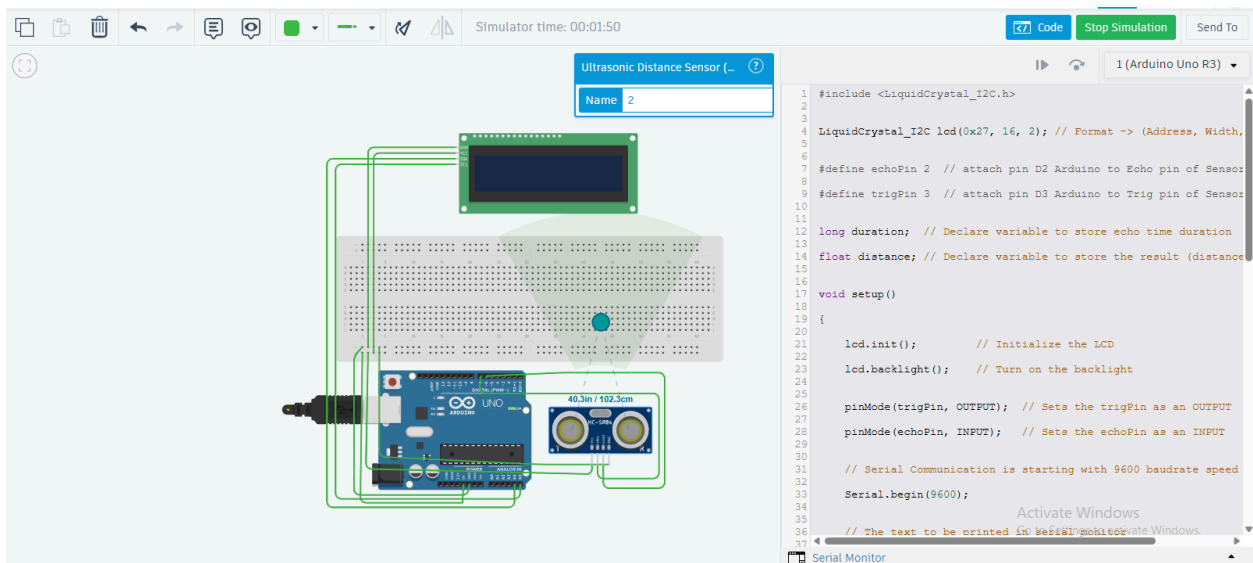
// Display on LCD
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Distance:");
lcd.setCursor(0, 1);
lcd.print(distance);
lcd.print(" cm");

// Print to Serial Monitor
Serial.print("Distance: ");
Serial.print(distance);
Serial.println(" cm");

delay(500);
}
```

Step 5: Start the Simulation

1. Click the "Start Simulation" button.
2. Observe the LCD displaying the measured distance in centimeters.
3. If the LCD is blank, adjust the potentiometer to change the contrast.
4. Open the Serial Monitor (Click on "Code" → "Monitor") to check values.



Conclusion:

The simulation successfully demonstrated the distance measurement using an ultrasonic sensor in Tinkercad. The measured distance was accurately displayed on a 16x2 LCD screen, showcasing how ultrasonic sensors can be effectively utilized in various applications such as automation, robotics, and obstacle detection.

This experiment provided valuable insights into:

- The process of interfacing an ultrasonic sensor with an Arduino board.
- The method of displaying sensor data on an LCD screen for real-time feedback.
- The advantages of simulating and testing circuits in Tinkercad before implementing them in the real world, ensuring accuracy and reducing the risk of errors during physical construction.

Overall, the simulation served as an excellent tool to understand and visualize the working of ultrasonic sensors, laying the groundwork for future real-world applications.